

Muhammad Rizwan

Software Engineer | Applied AI, RAG & Agentic Systems

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PROFESSIONAL SUMMARY

Software engineer with 5+ years shipping payroll, HR, and payments products end to end, now building the AI side of them. Works in Python with RAG pipelines, semantic search, vector databases, and agentic, tool-using workflows, and knows how to put a model behind a real API and interface. Holds the AWS ML Engineer Nanodegree.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL

Applied AI: RAG, retrieval pipelines, grounded generation, agentic workflows, LLMs, transformers, semantic search, vector databases, embeddings, retrieval evaluation, tool use, reflection, planning, multi-agent workflows, error analysis

ML & NLP: TensorFlow, PyTorch, scikit-learn, NumPy, NLP, word embeddings, sequence models, attention, encoder-decoder models, summarization, question answering, supervised learning, unsupervised learning, model evaluation

AWS & Deployment: Amazon SageMaker, AWS Lambda, Amazon S3, Amazon EC2, AWS Step Functions, model deployment, SageMaker endpoints

Hugging Face: Transformers, Datasets, Tokenizers, Accelerate, tokenization, model inference, fine-tuning

AI-Assisted Development: Claude, Claude Code, OpenAI Codex, Gemini, Cursor, AI pair programming, agentic coding workflows, prompt engineering, AI-assisted code review

Full-Stack Engineering: React, Vue.js, Node.js, Express.js, ASP.NET Core, REST APIs, PostgreSQL, Docker

PROJECTS

Skin Cancer Detection (Kaggle ISIC 2024) | Udacity capstone | mrizzy.vercel.app/work/isic-skin-cancer-detection

- Built a model that ranks a patient's skin lesions by cancer risk on 401,059 lesions with only 393 malignant, blending LightGBM on lesion metadata with a ResNet18 CNN on image crops.
- Reached 0.169 partial AUC (maximum 0.2) in patient-grouped 5-fold cross-validation, up from 0.112 for a logistic regression benchmark, and 0.159 on Kaggle's private test set as a late submission.
- Trained, tuned (12-run tuning job), and deployed the model to an Amazon SageMaker endpoint.

PROFESSIONAL EXPERIENCE

Frontend Engineer | Worklio

Nov 2022 - Present

- Developed WageTime and Worklio's embedded, white-label payroll and HR solutions across React and Vue frontends, shared TypeScript SDK and UI packages, and C#/.NET APIs.
- Built complex scheduling workflows covering recurring schedules, employee and team assignments, daily, weekly, and monthly calendars, schedule changes, conflict review, and time-off-aware coverage.
- Contributed to ASP.NET Core REST APIs and supporting API-model and business layers, coordinating frontend and backend contracts across WageTime modules.

Technical Team Lead | GitStart

Apr 2022 - Nov 2022

Full-Stack Software Engineer | GitStart

Oct 2021 - Apr 2022

- Architected backend services with Node.js, Express.js, and PostgreSQL for scalable, reliable web products.
- Built React and Next.js interfaces and integrated them with backend APIs across client and open-source projects.
- Served as the primary code reviewer for a distributed team, enforcing quality standards and guiding implementation decisions.

Full-Stack Software Engineer | Avybe

Aug 2020 - Jul 2021

- Designed secure, scalable backend APIs and integrated them with frontend workflows.
- Integrated PayPal and Stripe using webhooks for real-time transaction updates and payment processing.
- Reduced load times in an existing mobile application by 30% through code refactoring and performance tuning.

CERTIFICATIONS & TRAINING

AWS Machine Learning Engineer Nanodegree | Udacity

Software Architect Nanodegree | Udacity

Agentic AI | DeepLearning.AI

Retrieval Augmented Generation (RAG) | DeepLearning.AI

Natural Language Processing Specialization | DeepLearning.AI

LLM Course | Hugging Face

Deep Learning Specialization | DeepLearning.AI